Week 5 cell counting check-in

Ruiping Huang | RGB and grayscale preprocessing comparison

Result. RGB achieved lower total-cell counting error than grayscale in this seed-0 YOLO26n experiment. Grayscale did not improve the tested configuration.

Question and preparation. The team EDA raised background color as a possible confounder. I tested grayscale conversion using 500 Roboflow Version 1 originals, with the team's frozen 351/74/75 train/validation/test membership. All 26,715 Live and Dead boxes were retained as one Cell class. Grayscale luminance was replicated into three channels; image geometry and paired labels were unchanged.

Training and evaluation. Both models started from the same official pretrained checkpoint, using 640-pixel inputs, batch 16, seed 0, horizontal/vertical flips, and right-angle rotations. Other random augmentation was disabled. Both used a 100-epoch maximum and patience 20 on a Colab T4. Existing validation-selected confidence thresholds, RGB 0.44 and grayscale 0.33, were applied to the same 75 test images.

Test metric	RGB	Grayscale
Count MAE, cells/image	1.173	1.587
Count RMSE, cells/image	1.592	2.248
R-squared	0.9885	0.9770
Within 10% of reference	75/75	71/75
Largest absolute count error	4	9

The within-10% measure is a tolerance rate. Exact totals occurred on 22 RGB and 19 grayscale images.

What worked and what did not. Both models counted all 56 cells in one reviewed example. Grayscale had higher MAE in three acquisition sources and equal MAE in two; the largest gap was source 3_A, 0.733 versus 1.933. The largest grayscale error was reference 65, RGB 61, grayscale 56. An RGB-exact case still had unmatched reference and prediction boxes that canceled in the total, showing why count metrics need visual diagnostics.

Limitations and next step. RGB completed 100 epochs; grayscale stopped at 39, with its best checkpoint from 19. Checkpoints and stopping used validation detection mAP50-95, not count MAE. This is one seed with all sources represented in every split. Post-hoc examples do not establish a color mechanism. I propose reviewing ambiguous/border annotations and agreeing on detection evaluation, then choosing repeated seeds or an equal-budget study. No supplementary training has started.

Team context. This retains the team's YOLO26n architecture but changes the frozen split, augmentation, stopping patience, and count evaluation relative to the inspected team notebook. It is not a direct replication. The project Word document reports historical Cellpose MAE 5.6; that run was not reproduced here, and its evaluation compatibility requires checking before a formal comparison.

Evidence: full report, source and image diagnostics, and team notebook comparison.